Electric Charge

1 Quantisation

$$q = \pm ne, \qquad e = 1.6 \times 10^{-19}\,\mathrm{C}$$

$n = 0, 1, 2, \ldots$ · $1\,\mathrm{C} = 1\,\mathrm{A\,s}$ · $1\,\mathrm{C}$ *is the charge of* 6.25×10^{18} *electrons*

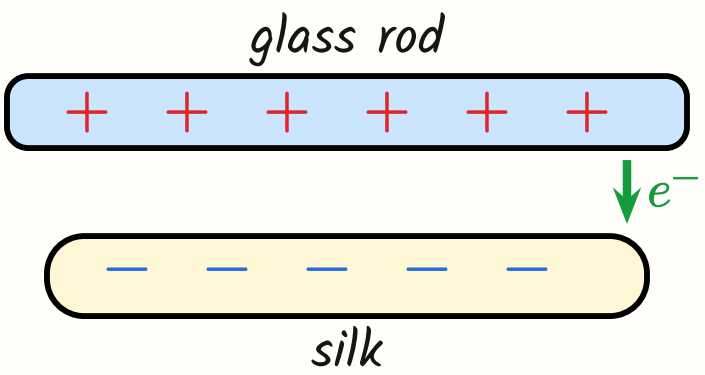
Example

$q = -3.2\,\mu\mathrm{C}$: $n = |q|/e = 2 \times 10^{13}$ *extra electrons.*

2 Other properties

- *Two kinds: like charges repel, unlike attract.*
- *Additive (keep signs):* $+2, -5, +1\,\mu\mathrm{C} \Rightarrow -2\,\mu\mathrm{C}$.
- *Conserved in an ISOLATED system: only transferred.*
- *Invariant: does not depend on the body's speed.*

3 Charging by rubbing



- *Electrons go from glass to silk.*
- *Glass* $+q$, *silk* $-q$ *(equal).*

Golden points

★ $2.4 \times 10^{-19}\,\mathrm{C}$ *is impossible* $(= 1.5\,e)$.

★ *Quarks* $(\pm e/3,\ \pm 2e/3)$ *are never found free.*

Summary

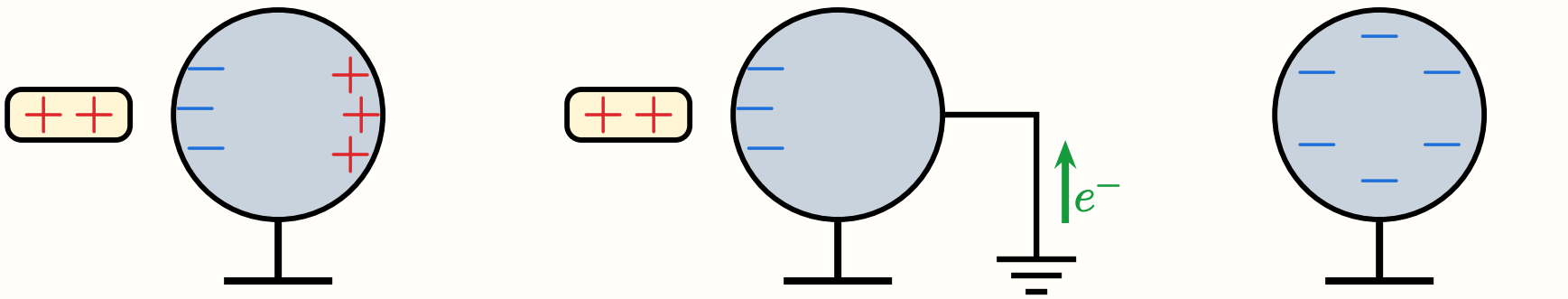
$q = ne$ • *additive* • *conserved* • *invariant*

Conduction and Induction

1 Conductors and insulators

- Conductors (metals, earth): free electrons. Insulators (glass): charge stays put.
- Earthing: charge flows to the huge earth.

2 Charging by induction (no contact)



(1) rod near (2) earth it (3) remove earth, then the rod

- Induced charge is opposite to the rod's; the rod loses nothing.
- Break the earth link BEFORE taking the rod away.

3 Charging by conduction (contact)

Example

Identical metal spheres $+6\,\mu\text{C}$ and $-2\,\mu\text{C}$ touched: each gets $\dfrac{6+(-2)}{2} = +2\,\mu\text{C}$.

Golden points

★ A charged body also attracts a NEUTRAL one (induction): only repulsion proves charge.

★ Equal sharing only for IDENTICAL conductors.

Coulomb's Law

1 Two point charges

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2} = k\frac{|q_1 q_2|}{r^2}$$

$k = 9 \times 10^9\,\mathrm{N\,m^2\,C^{-2}}$ · $\varepsilon_0 = 8.854 \times 10^{-12}\,\mathrm{C^2\,N^{-1}\,m^{-2}}$ · *point charges at rest, in vacuum*

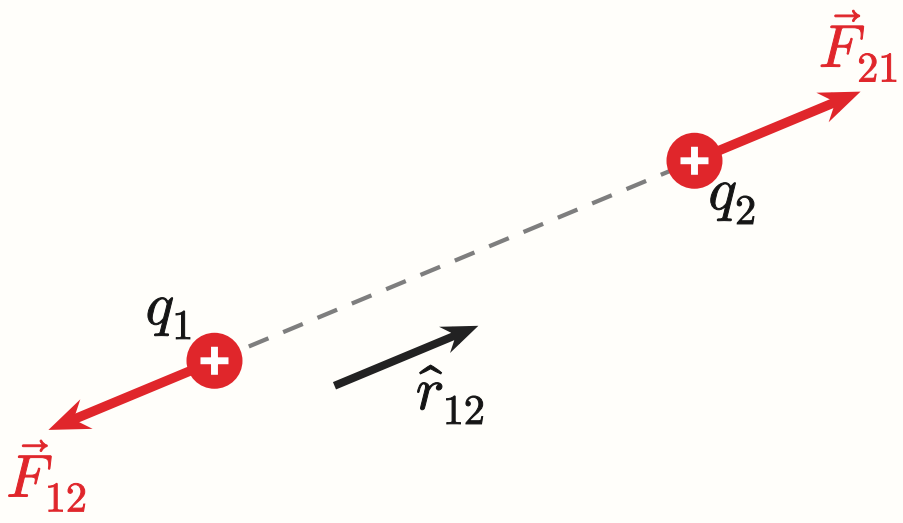
Example

$2\,\mu\mathrm{C}$ *and* $-3\,\mu\mathrm{C}$, $0.3\,\mathrm{m}$ *apart:* $F = 0.6\,\mathrm{N}$, *attractive.*

2 Vector form

$$\vec{F}_{21} = \frac{k\,q_1 q_2}{|\vec{r}_2 - \vec{r}_1|^3}(\vec{r}_2 - \vec{r}_1)$$

force ON q_2 *BY* q_1 · *signs included:* $q_1 q_2 > 0$ *repels*



- $\vec{F}_{12} = -\vec{F}_{21}$ *(3rd law).*
- *Along the joining line (central).*
- *Drawn: both +, they repel.*

Golden points

★ *Spheres: centre-to-centre distance (spherical charge only); close conductors redistribute charge.*

★ *Air:* $\varepsilon_r \approx 1.0006$, *so use the vacuum value of* k.

Coulomb's Law in a Medium

1 Dielectric fills all the space

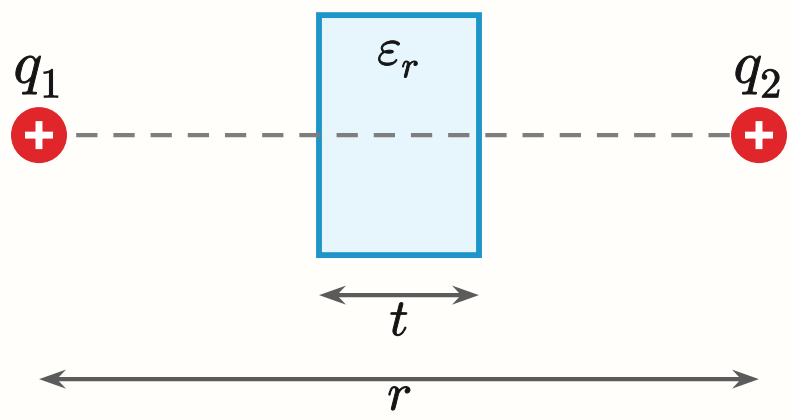
$$F_m = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{|q_1 q_2|}{r^2} = \frac{F_{vac}}{\varepsilon_r}$$

$\varepsilon = \varepsilon_0\varepsilon_r$ · $\varepsilon_r = K \geq 1$, *no unit* · *medium fills all space around the charges*

Example

The $0.6\,\mathrm{N}$ *pair, in water* ($\varepsilon_r = 80$): $F_m = 0.6/80 = 7.5 \times 10^{-3}\,\mathrm{N}$.

2 Slab of thickness t in the gap



$$F = \frac{k\,|q_1 q_2|}{\left(r - t + t\sqrt{\varepsilon_r}\right)^2}$$

slab fills the gap, faces $\perp$ *the line*

Example

$r = 0.5,\ t = 0.1\,\mathrm{m},\ \varepsilon_r = 4$:

$r_{eff} = 0.6\,\mathrm{m}$.

3 Electric vs gravitational force

- *Electron–proton:* $F_e/F_g = ke^2/Gm_e m_p \approx 2.3 \times 10^{39}$, *at every* r.
- *Gravity only attracts and is the same in any medium.*

Golden points

★ $[\varepsilon_0] = [\mathrm{M}^{-1}\mathrm{L}^{-3}\mathrm{T}^4\mathrm{A}^2]$; *slab only partly filling: NOT* F/ε_r.

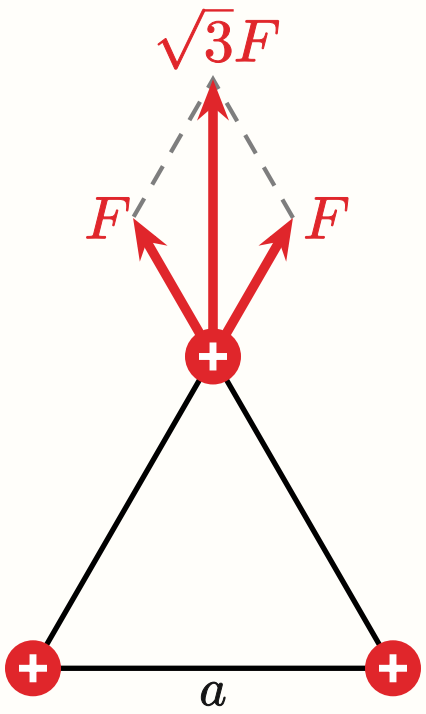
Superposition of Forces

1 Vector sum of pair forces

$$\vec{F}_q = \vec{F}_{q1} + \vec{F}_{q2} + \cdots + \vec{F}_{qn}$$

each pair force as if the others were absent (point charges, fixed in place)

2 Equilateral triangle, three equal charges q



- Each force $F = kq^2/a^2$, 60° apart.
- Resultant $\sqrt{3}\,kq^2/a^2$, along the outward bisector (to scale).

Example

$q = 1\,\mu\text{C}$, $a = 0.1\,\text{m}$: $F = 0.9\,\text{N}$, *net* $1.56\,\text{N}$.

3 Square of side a, four equal charges q

$$F_{corner} = \frac{kq^2}{a^2}\left(\sqrt{2} + \frac{1}{2}\right) \approx 1.91\,\frac{kq^2}{a^2}$$

along the diagonal, away from the centre

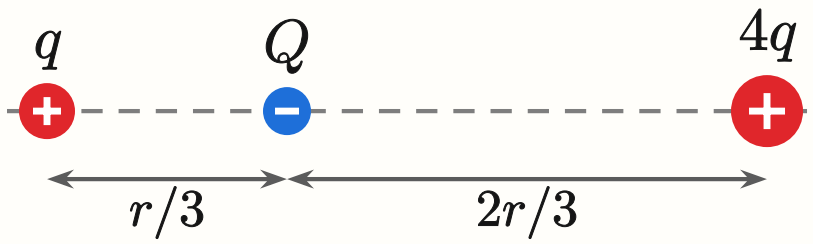
Golden points

★ Add VECTORS, not magnitudes; let symmetry cancel components.

★ Centre of a symmetric set of equal charges: net force zero.

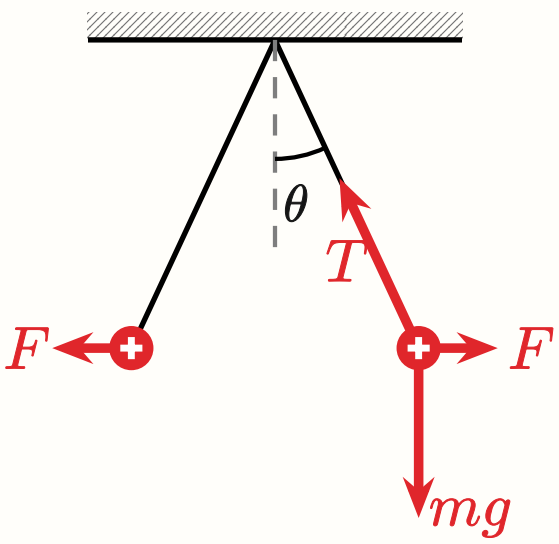
Equilibrium of Charges

1 Free third charge Q on the line of q_1, q_2



- Like pair: Q between, nearer the smaller q_1: $x = \dfrac{r}{1+\sqrt{q_2/q_1}}$ from q_1.
- Unlike pair: outside, beyond the smaller: $x = \dfrac{r}{\sqrt{|q_2/q_1|}-1}$.
- $q, 4q$: $x = r/3$; all three rest only if $Q = -\dfrac{q_1 q_2}{(\sqrt{q_1}+\sqrt{q_2})^2} = -\dfrac{4}{9}q$.
- Earnshaw: no STABLE rest under electrostatic forces alone.

2 Suspended charged balls



$$\tan\theta = \frac{F}{mg}$$

$F = \dfrac{kq^2}{(2L\sin\theta)^2}$, L: string length

- In a liquid (density σ; balls ρ), θ unchanged if $\varepsilon_r = \dfrac{\rho}{\rho-\sigma}$.
- Example: $\rho = 1600$, $\sigma = 800\,\mathrm{kg\,m^{-3}}$; $\varepsilon_r = 2$.

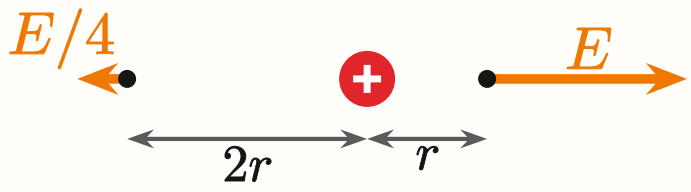
Electric Field

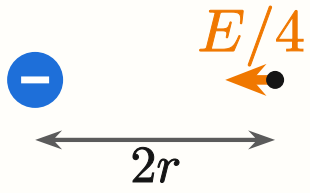
1 Force per unit test charge

$$\vec{E} = \lim_{q_0 \to 0} \frac{\vec{F}}{q_0}, \qquad \vec{E} = \frac{kq}{r^2}\,\hat{r}$$

$\mathrm{N\,C^{-1}} = \mathrm{V\,m^{-1}}$ · $[\mathrm{MLT^{-3}A^{-1}}]$ · $\hat{r}$: source to point · vacuum

2 Direction and fall-off





- $+q$: $\vec{E}$ away; $-q$: towards it.
- $\vec{F} = q\vec{E}$: along $\vec{E}$ on $+q$, opposite on $-q$.
- $E \propto 1/r^2$: at $2r$ it is $E/4$ (to scale).

Example

$q = 4\,\mathrm{nC}$, $r = 0.3\,\mathrm{m}$: $E = 400\,\mathrm{N\,C^{-1}}$; on an electron there $F = 6.4 \times 10^{-17}\,\mathrm{N}$, towards q.

3 Many charges

$$\vec{E} = \sum_i \frac{kq_i}{r_i^2}\,\hat{r}_i$$

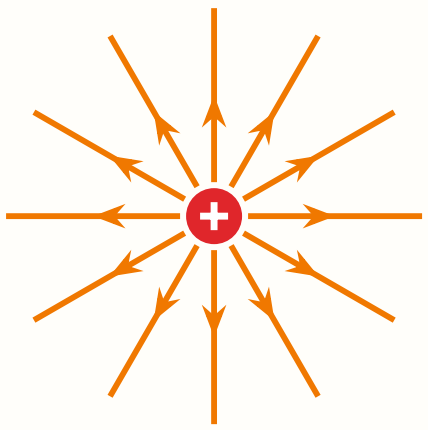
neutral point ($\vec{E} = 0$) of q, $4q$ (distance r): $r/3$ from q

Golden points

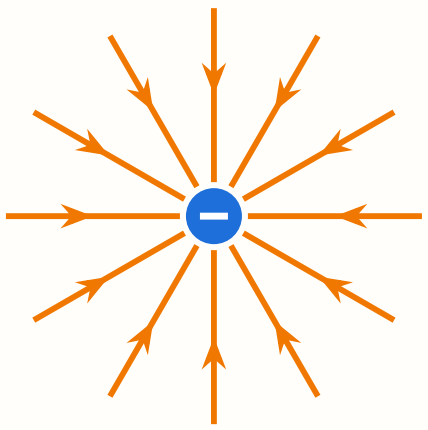
★ $\vec{E}$ exists with or without a test charge; $q_0 \to 0$ so the sources are not disturbed.

Electric Field Lines

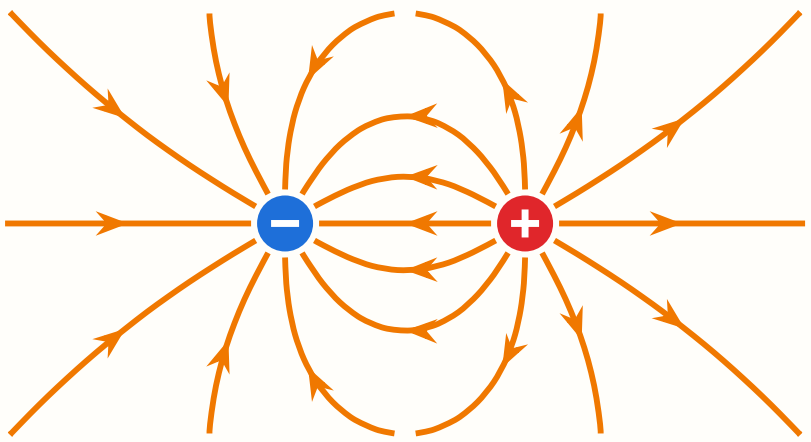
1 Patterns (traced from Coulomb's law)



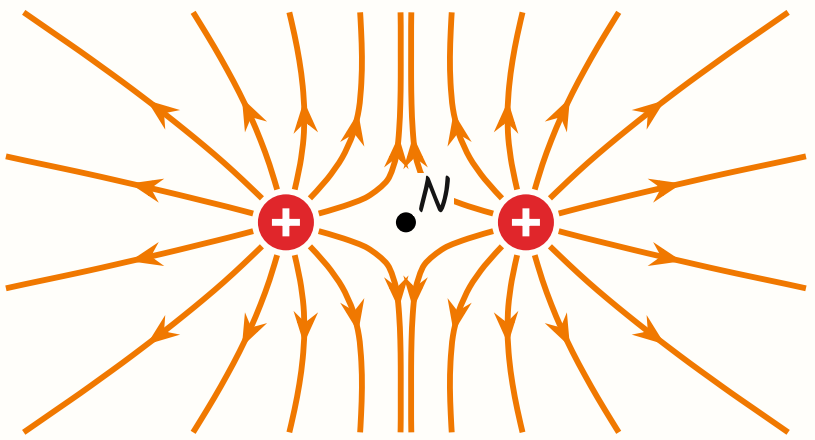
single $+q$: out



single $-q$: in



dipole: from $+q$ to $-q$



$+q$, $+q$: neutral point N

2 Properties

- Start on $+$, end on $-$ (or at infinity).
- Tangent = direction of $\vec{E}$; lines never cross.
- Crowded lines = strong field; number of lines $\propto q$.
- No closed loops (electrostatic field).
- Meet a conductor at 90° (in equilibrium).

Golden points

★ A field line is NOT the path of a moving charge.

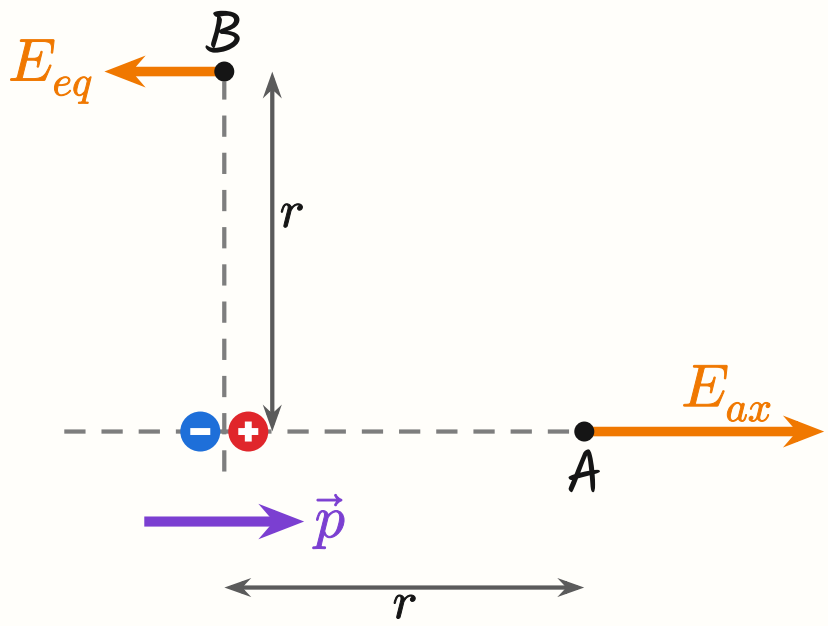
Electric Dipole

1 Dipole moment

$$\vec{p} = q\,(2\vec{a}), \qquad \text{from } -q \text{ to } +q$$

$2a$: separation · unit C m · total charge zero

2 Axial point A and equatorial point B



- $\vec{E}_{ax}$ along $\vec{p}$; $\vec{E}_{eq}$ opposite to $\vec{p}$.
- Same $r \gg a$: $E_{ax} = 2E_{eq}$ (drawn $2:1$).
- r measured from the centre.

$$E_{ax} = \frac{2kp\,r}{(r^2 - a^2)^2} \approx \frac{2kp}{r^3}$$

$$E_{eq} = \frac{kp}{(r^2 + a^2)^{3/2}} \approx \frac{kp}{r^3}$$

exact for any $r > a$; $\approx$ only far away ($r \gg a$)

Example

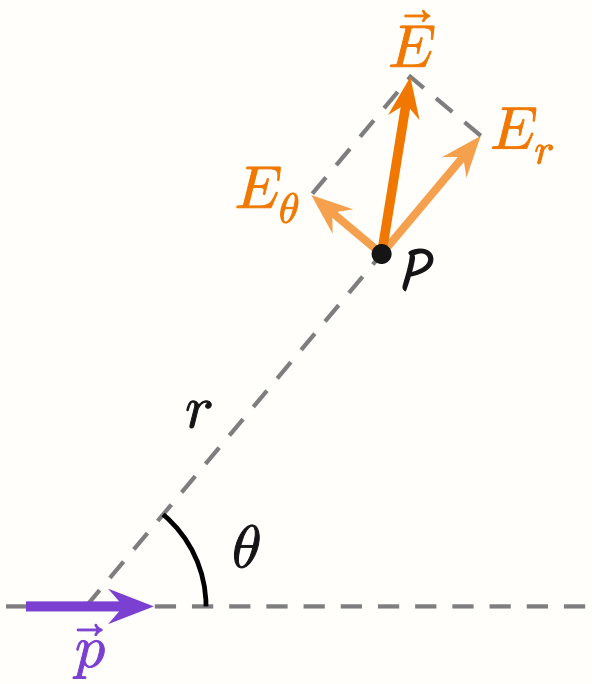
$p = 2 \times 10^{-8}\,\mathrm{C\,m}$, $r = 0.1\,\mathrm{m} \gg a$: $E_{ax} = 3.6 \times 10^5$, $E_{eq} = 1.8 \times 10^5\,\mathrm{N\,C^{-1}}$.

Golden points

★ Far field $\propto 1/r^3$; ideal dipole: $2a \to 0$ with p finite.

Dipole Field at Any Point

1 Point at (r, θ), θ from $\vec{p}$



$$E_r = \frac{2kp\cos\theta}{r^3}$$

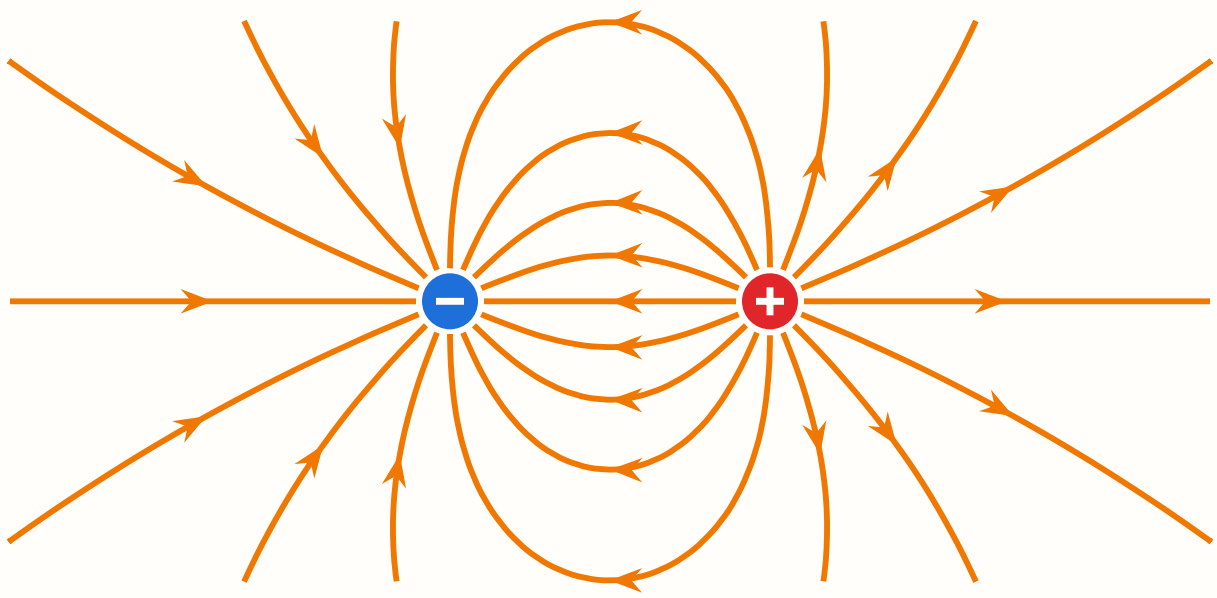
$$E_\theta = \frac{kp\sin\theta}{r^3}$$

$$E = \frac{kp}{r^3}\sqrt{1+3\cos^2\theta}$$

$r \gg a$

- Direction: $\tan\alpha = \frac{1}{2}\tan\theta$, α = angle of $\vec{E}$ from $\hat{r}$.
- $\vec{E} \perp \vec{p}$ where $\tan\theta = \sqrt{2}$ ($\theta \approx 54.7°$). Drawn to scale, $\theta = 50°$.

2 Field pattern of a dipole (traced)



Golden points

★ Between the charges $\vec{E}$ points from $+$ to $-$: opposite to $\vec{p}$.

★ Polar molecules (H_2O) have a permanent $\vec{p}$.

Dipole in a Uniform Field

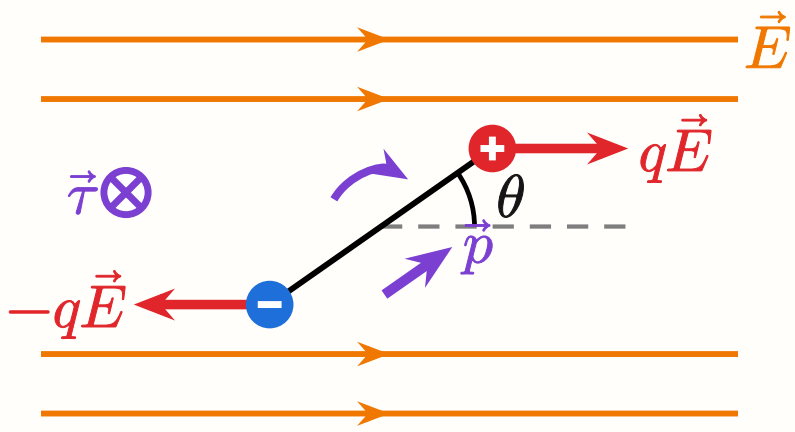
1 Torque, force, energy

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE\sin\theta, \quad \vec{F}_{net} = 0$$

$$U = -\vec{p}\cdot\vec{E} = -pE\cos\theta$$

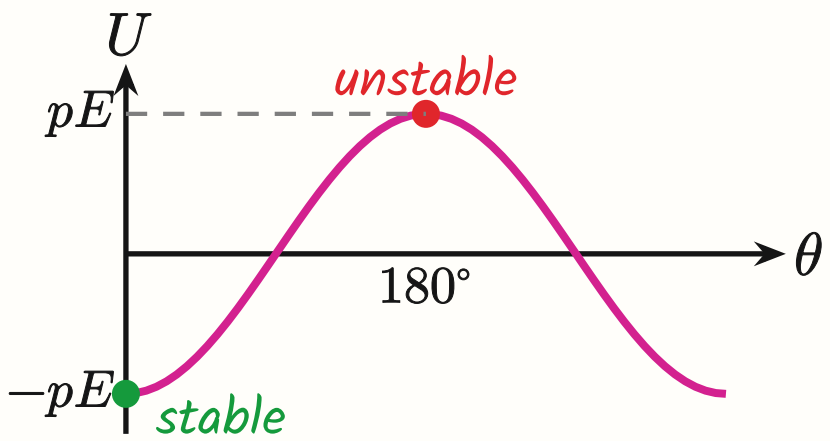
UNIFORM $\vec{E}$ · θ: from $\vec{E}$ to $\vec{p}$ · $U = 0$ chosen at $\theta = 90°$

2 Couple on the dipole



- Equal, opposite forces: net force zero, but a couple.
- It turns $\vec{p}$ towards $\vec{E}$ (here clockwise: $\vec{\tau}$ into the page).

3 Energy and equilibrium



- $\theta = 0$: stable, $U = -pE$ (min).
- $\theta = 180°$: unstable, $U = +pE$.
- $W_{\theta_1\to\theta_2} = pE(\cos\theta_1 - \cos\theta_2)$.

Example

$p = 4\times10^{-9}\,\mathrm{C\,m}$, $E = 5\times10^{4}\,\mathrm{N\,C^{-1}}$: $\tau(30°) = 10^{-4}\,\mathrm{N\,m}$;
$W_{0\to180°} = 2pE = 4\times10^{-4}\,\mathrm{J}$.

Golden points

★ NON-uniform $\vec{E}$: the net force on a dipole is NOT zero.

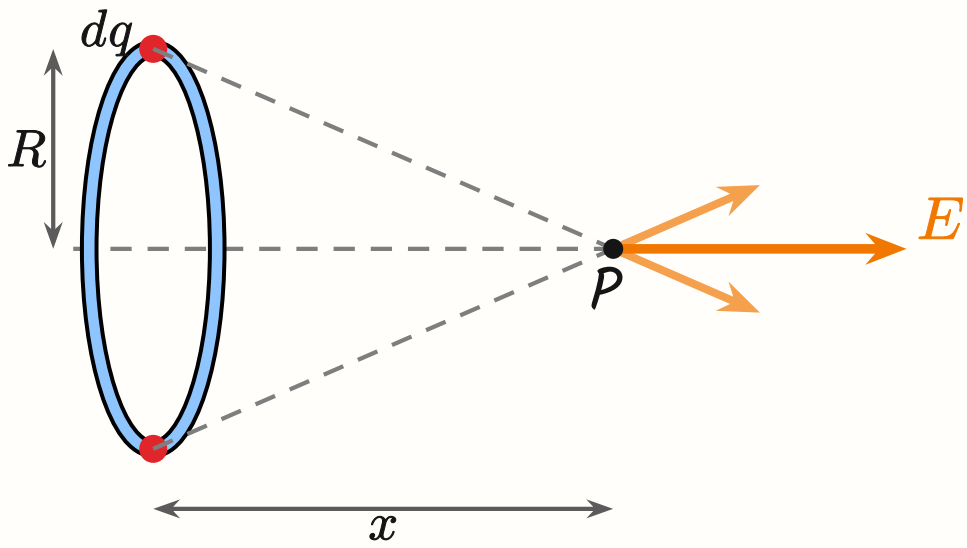
Continuous Charge and Ring

1 Charge densities

$$\lambda = \frac{dq}{dl}, \quad \sigma = \frac{dq}{dA}, \quad \rho = \frac{dq}{dV}, \qquad \vec{E} = \int \frac{k\,dq}{r^2}\,\hat{r}$$

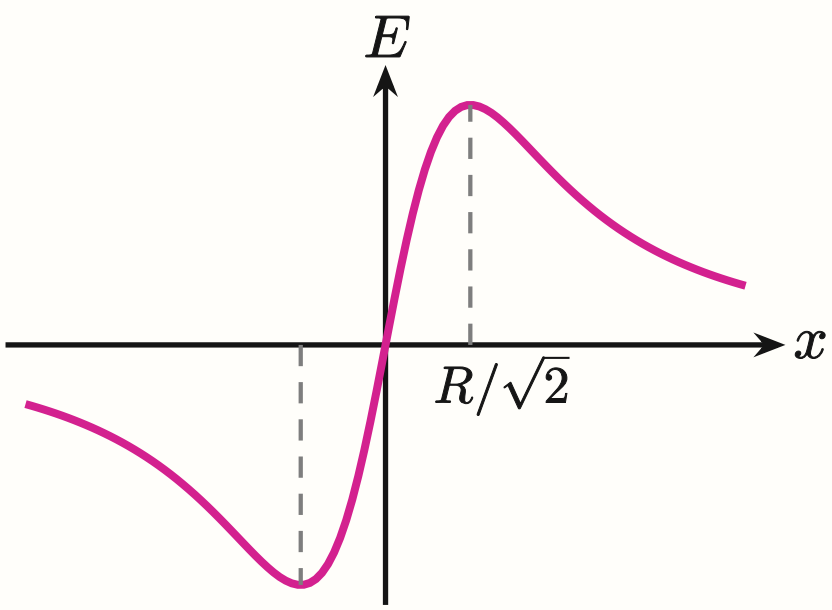
units: $C\,m^{-1}$, $C\,m^{-2}$, $C\,m^{-3}$ · *add the VECTOR* $d\vec{E}$ *(use symmetry)*

2 Uniform ring (charge Q) on its axis



$$E = \frac{kQ\,x}{(R^2 + x^2)^{3/2}}$$

along the axis, away from a + ring
· *⊥ parts cancel in pairs*



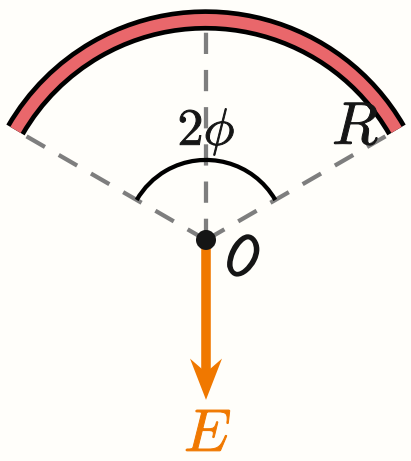
- *Centre* ($x = 0$): $E = 0$.
- *Max at* $x = \pm R/\sqrt{2}$:
  $E_{max} = \dfrac{2kQ}{3\sqrt{3}R^2}$.
- $x \gg R$: $E \approx kQ/x^2$ *(point charge).*

Golden points

★ *A* $-q$ *released on the axis near the centre* ($x \ll R$) *of a* $+Q$ *ring does SHM:* $\omega^2 = \dfrac{kQq}{mR^3}$.

Arc and Line Charge

1 Arc: radius R, angle 2ϕ at the centre

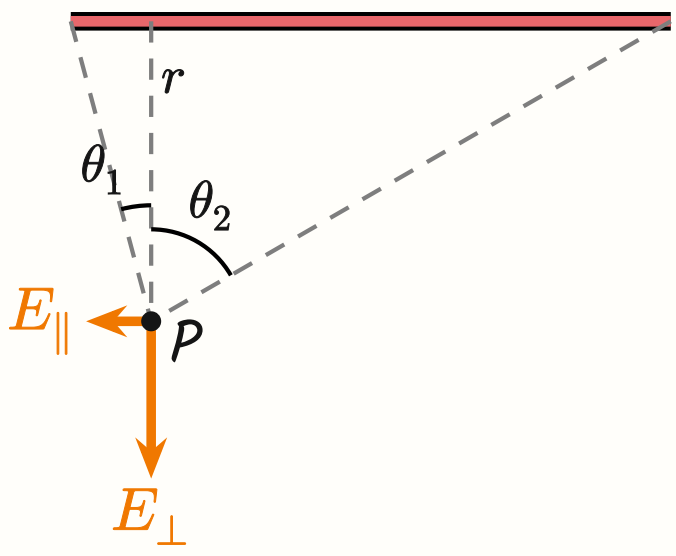


$$E_O = \frac{2k\lambda \sin\phi}{R}$$

uniform λ · along the bisector, away from a + arc

- *Semicircle:* $2k\lambda/R$; *full ring:* 0.

2 Straight line charge, point P at distance r



$$E_\perp = \frac{k\lambda}{r}(\sin\theta_1 + \sin\theta_2)$$

$$E_\parallel = \frac{k\lambda}{r}(\cos\theta_1 - \cos\theta_2)$$

angles from the perpendicular, ends on opposite sides · $E_\parallel$ points away from the longer part

- *Drawn to scale:* $\theta_1 = 15°$, $\theta_2 = 60°$.
- *Infinite line* ($\theta_1 = \theta_2 = 90°$): $E = \dfrac{2k\lambda}{r} = \dfrac{\lambda}{2\pi\varepsilon_0 r}$, $E_\parallel = 0$.
- *Semi-infinite, P opposite one end:* $E_\perp = E_\parallel = k\lambda/r$, *at* $45°$.

Summary

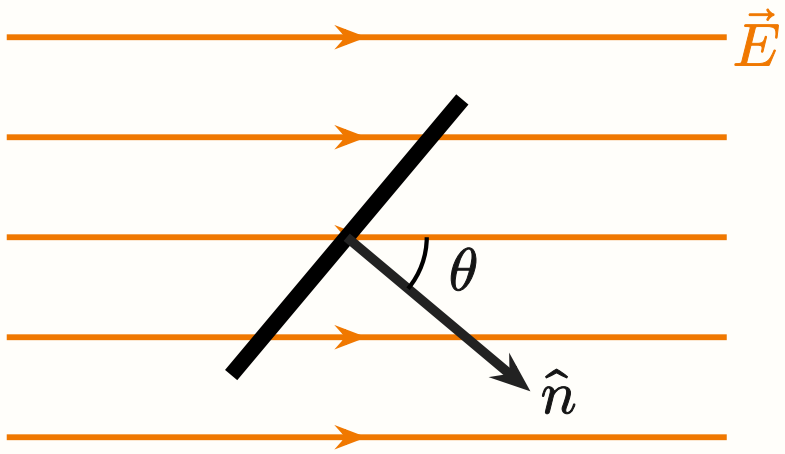
arc $2k\lambda \sin\phi/R$ • *infinite line* $2k\lambda/r$

Electric Flux

1 Flux through an area

$$\Phi = \vec{E}\cdot\vec{A} = EA\cos\theta, \qquad \Phi = \int \vec{E}\cdot d\vec{A}$$

$\vec{A}$ along the normal (OUTWARD on a closed surface) · scalar · $\mathrm{N\,m^2\,C^{-1}}$



- Plane seen edge-on; θ is from the NORMAL.
- $\theta = 90°$: $\Phi = 0$.
- Lines out: $+$; lines in: $-$.

Example

$\vec{E} = (3\hat{i} + 4\hat{j}) \times 10^3\,\mathrm{N\,C^{-1}}$, $\vec{A} = 0.02\,\hat{i}\,\mathrm{m^2}$: $\Phi = 60\,\mathrm{N\,m^2\,C^{-1}}$.

2 Solid angle

$$\Omega = \frac{A\cos\theta}{r^2}, \qquad \Phi = \frac{q\,\Omega}{4\pi\varepsilon_0}$$

steradian, whole sphere 4π · point charge q · disc at half-angle α: $\Omega = 2\pi(1-\cos\alpha)$

- Disc with $R = \sqrt{3}\,d$ ($\alpha = 60°$): $\Omega = \pi$, $\Phi = q/4\varepsilon_0$.

Golden points

★ Uniform $\vec{E}$, any CLOSED surface: $\Phi = 0$.

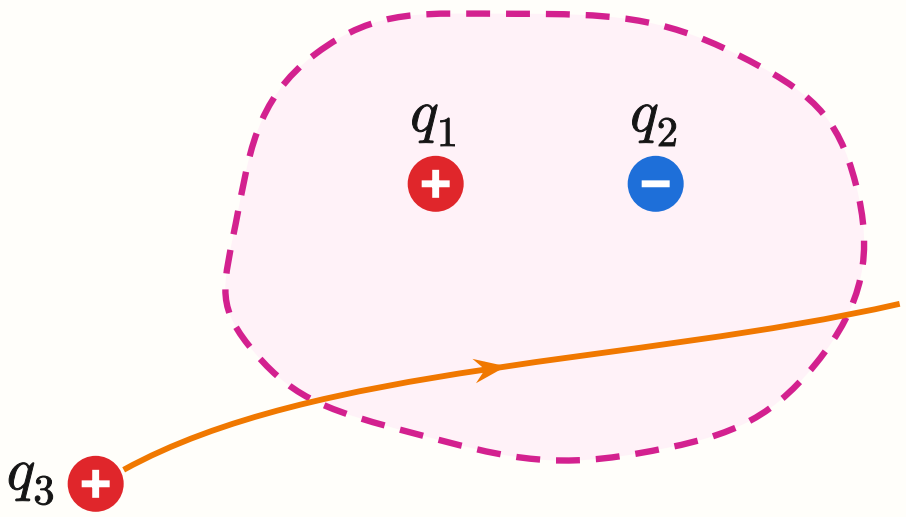
★ Hemisphere, uniform $\vec{E}$ along its axis: curved part $\pi R^2 E$.

Gauss's Law

1 Any closed surface

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0}$$

q_{enc}: net charge INSIDE · $\vec{E}$: TOTAL field of all charges, inside and outside · vacuum



- $\Phi = (q_1 + q_2)/\varepsilon_0$.
- q_3 outside: its lines go in and out: net flux 0, but it changes $\vec{E}$ on the surface.

Example

Inside $+2$ and $-5\,\mu\text{C}$, outside $+3\,\mu\text{C}$:

$$\Phi = \frac{-3 \times 10^{-6}}{8.854 \times 10^{-12}} = -3.39 \times 10^{5}\,\text{N}\,\text{m}^2\,\text{C}^{-1}.$$

2 Using it to find E

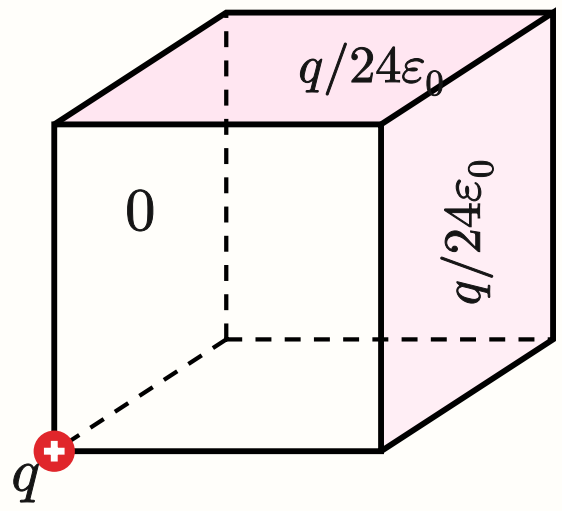
- ALWAYS true, but gives E easily only with symmetry: sphere, infinite line, infinite sheet.
- On the chosen surface E is constant and $\perp$ it, or $\parallel$ it.
- The surface must not pass through a point charge.

Golden points

★ Flux depends only on q_{enc}: not on the shape, size or where the charge sits inside.

Flux Through a Cube

1 Charge q at a corner



- 8 such cubes surround the corner: this cube gets $q/8\varepsilon_0$.
- 3 faces touching q (front, left, bottom): 0, $\vec{E}$ lies in them.
- 3 far faces (top, right, back): $q/24\varepsilon_0$ each.

2 Shares by position (symmetry)

Where q sits	Flux through cube
Centre	q/ε_0 ($q/6\varepsilon_0$ per face)
Centre of a face	$q/2\varepsilon_0$
Middle of an edge	$q/4\varepsilon_0$
Corner	$q/8\varepsilon_0$

Example

Square of side a with q at $a/2$ above its centre: imagine the cube of side a around q: $\Phi = q/6\varepsilon_0$.

Golden points

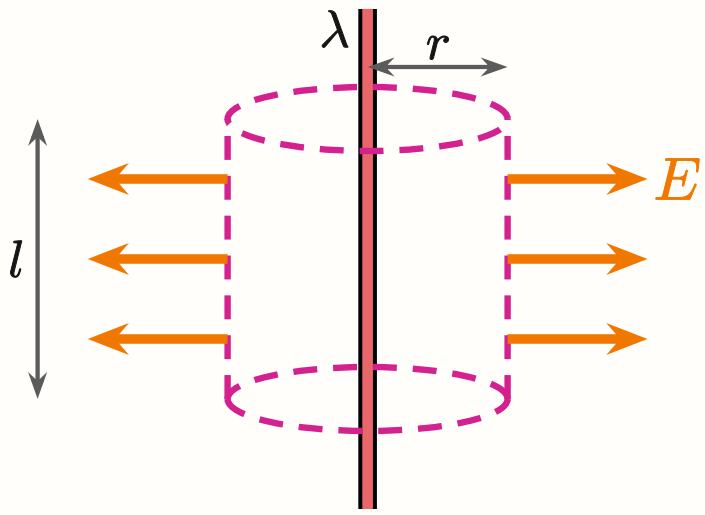
★ For q ON a face, edge or corner, count how many identical cubes would share it (2, 4 or 8).

Summary

centre $\frac{q}{6\varepsilon_0}$/face • face $\frac{q}{2\varepsilon_0}$ • edge $\frac{q}{4\varepsilon_0}$ • corner $\frac{q}{8\varepsilon_0}$

Line Charge and Sheet

1 Infinite line, Gaussian cylinder

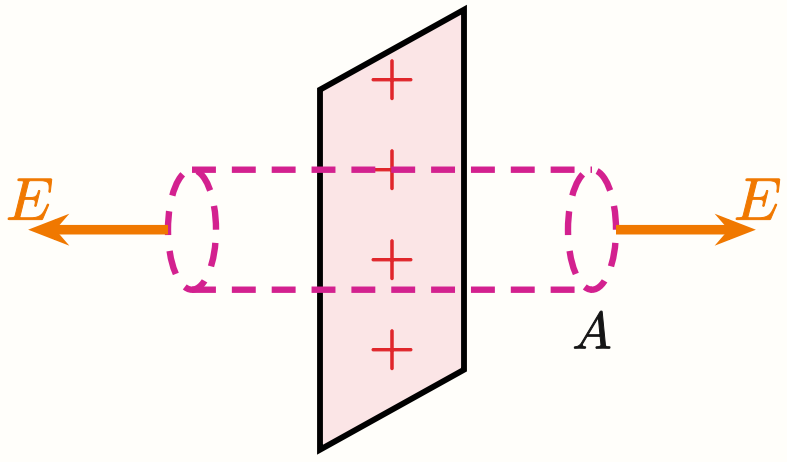


$$E\,(2\pi r l) = \frac{\lambda l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r}$$

infinitely long (ends far away) · radial · end caps: zero flux

2 Infinite plane sheet, Gaussian pillbox



$$E\,(2A) = \frac{\sigma A}{\varepsilon_0} \Rightarrow E = \frac{\sigma}{2\varepsilon_0}$$

infinite sheet · same at all distances · ⊥ sheet, away if $\sigma > 0$

3 Uniformly charged disc, on its axis

$$E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{x^2 + R^2}}\right)$$

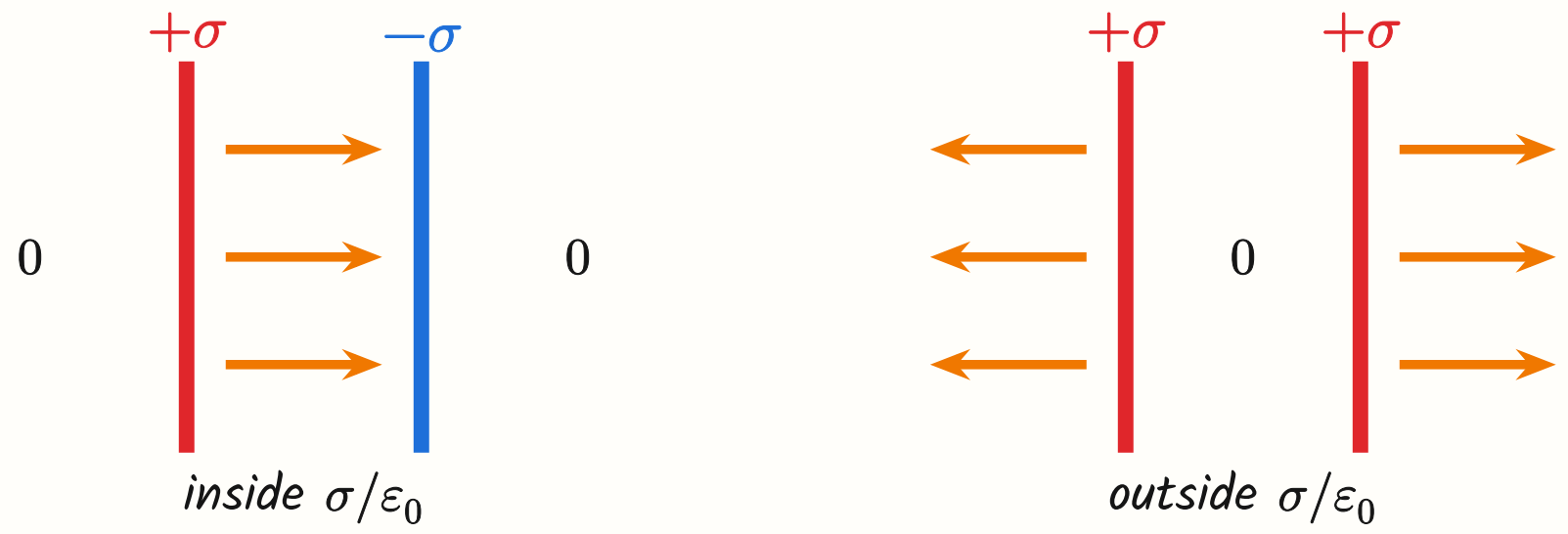
$R \gg x$: sheet value $\sigma/2\varepsilon_0$ · $x \gg R$: point charge kQ/x^2

Example

Infinite line, $\lambda = 2\,\mu\mathrm{C\,m^{-1}}$, $r = 0.2\,\mathrm{m}$:
$E = 2k\lambda/r = 1.8 \times 10^5\,\mathrm{N\,C^{-1}}$.

Two Parallel Sheets

1 Superpose $\sigma/2\varepsilon_0$ from each sheet



- *Infinite sheets (big plates, far from the edges). Arrow length $\propto E$: all arrows $= \sigma/\varepsilon_0$.*
- *$+\sigma, -\sigma$: add between, cancel outside (capacitor).*
- *$+\sigma, +\sigma$: cancel between, add outside.*

2 General: σ_1 (left), σ_2 (right)

$$E_{in} = \frac{\sigma_1 - \sigma_2}{2\varepsilon_0}, \qquad E_{out} = \frac{\sigma_1 + \sigma_2}{2\varepsilon_0}$$

signs of σ included · $E_{in} > 0$: to the right · $E_{out} > 0$: away from the pair

Example

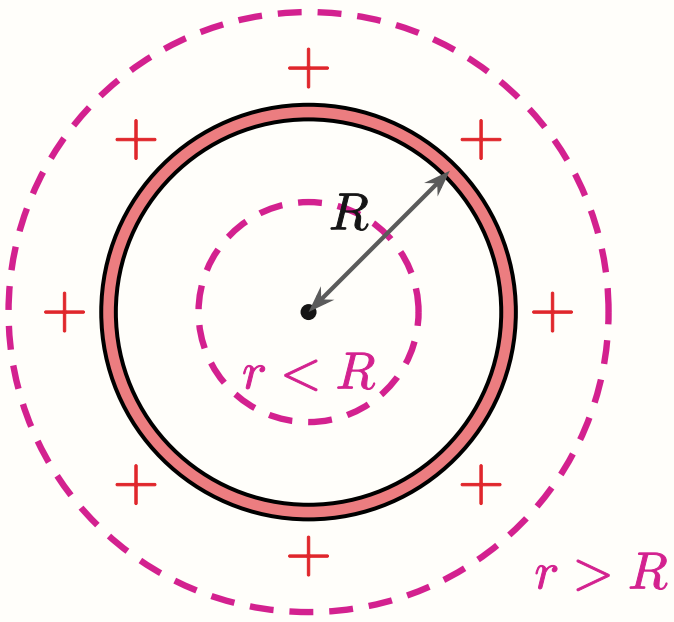
$\sigma_1 = +3\sigma_0$, $\sigma_2 = -\sigma_0$: *inside $2\sigma_0/\varepsilon_0$ (to the right); outside σ_0/ε_0 (away).*

Golden points

★ *A sheet's field does not fall with distance: the gap width does not change E.*

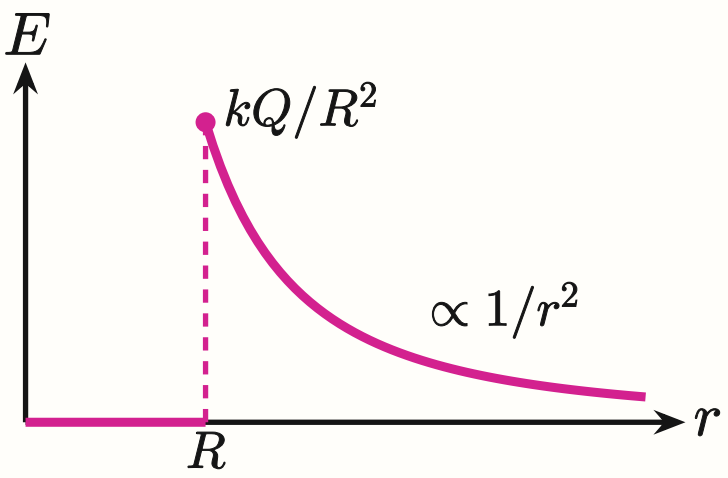
Spherical Shell

1 Thin shell, charge Q, radius R



$$E = \begin{cases} 0 & r < R \\ \dfrac{kQ}{r^2} & r > R \end{cases}$$

spherically symmetric charge; outside it acts as if all Q were at the centre



- *Inside:* zero ($q_{enc} = 0$).
- *Just outside:* $\dfrac{kQ}{R^2} = \dfrac{\sigma}{\varepsilon_0}$; *jumps at* $r = R$.

Example

$Q = 2\,\mu\text{C}$, $R = 0.1\,\text{m}$: $E(0.05) = 0$; *just outside* 1.8×10^6; $E(0.2) = 4.5 \times 10^5\,\text{N C}^{-1}$.

Golden points

★ *A charged CONDUCTING sphere (solid or hollow) gives the same graph: its charge sits on the surface.*

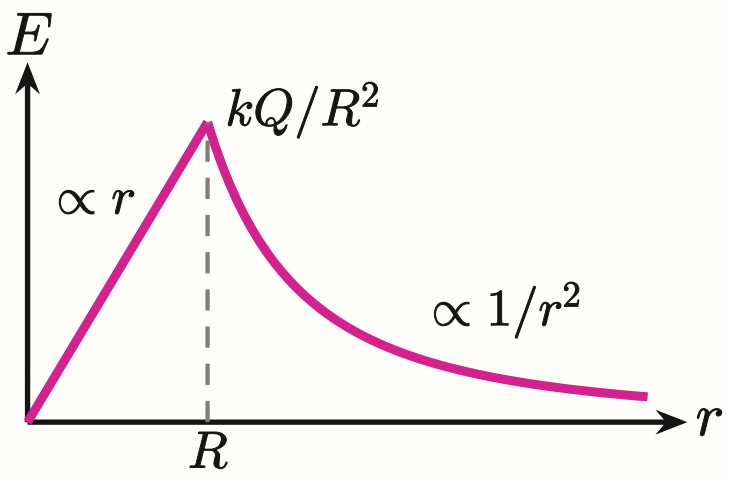
★ *Concentric shells: at r only the charge inside r counts. Inner $+Q$, outer $-Q$: kQ/r^2 between, 0 outside.*

Solid Sphere of Charge

1 Uniform ρ (non-conducting), radius R

$$E_{in} = \frac{kQ\,r}{R^3} = \frac{\rho\,r}{3\varepsilon_0}, \qquad E_{out} = \frac{kQ}{r^2}$$

uniform volume charge · $q_{enc} = Q\,r^3/R^3$ · E is largest at the surface



- *Linear inside, $1/r^2$ outside, continuous at R.*
- $E(R/2) = E(\sqrt{2}R) = \frac{1}{2}E(R)$.

2 Non-uniform: $\rho = \rho_0\, r/R$

$$q_{enc} = \frac{\pi\rho_0 r^4}{R}, \qquad E_{in} = \frac{\rho_0\, r^2}{4\varepsilon_0 R}$$

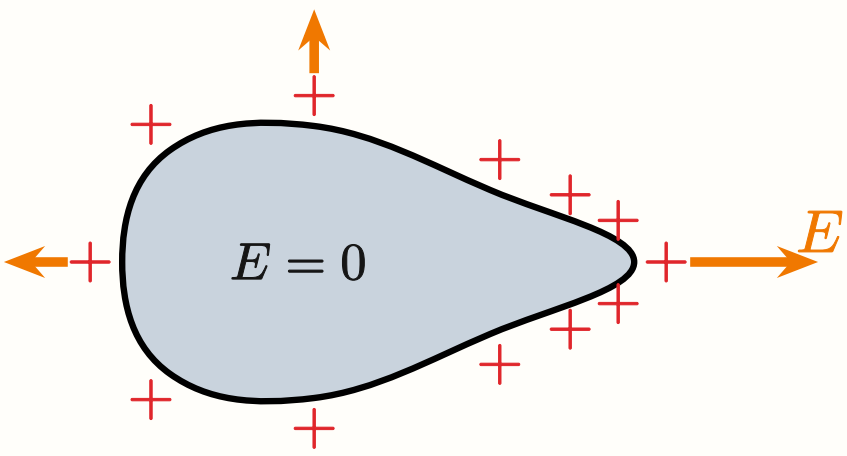
$r \le R$ · *use* $q_{enc} = \int_0^r \rho\, 4\pi r'^2 dr'$

3 How E falls with distance

point charge, sphere (outside)	$1/r^2$
infinite line charge	$1/r$
infinite sheet	*constant*
dipole (far, $r \gg a$)	$1/r^3$

Conductors in Equilibrium

1 Electrostatic equilibrium (no charge moving)



- $E = 0$ inside the metal.
- Net charge lives on the outer surface.
- $\vec{E}$ just outside is $\perp$ the surface.
- Sharper end: more σ, stronger E (longer arrow).

2 Field just outside a conductor

$$\vec{E} = \frac{\sigma}{\varepsilon_0}\,\hat{n}$$

σ: LOCAL density · $\hat{n}$: outward normal · electrostatic equilibrium

- Not $\sigma/2\varepsilon_0$: the rest of the charge doubles the patch's field outside (and cancels it inside).
- Far-apart spheres joined by a wire: $\sigma_1/\sigma_2 = R_2/R_1$.

3 Electrostatic pressure

$$P = \frac{\sigma^2}{2\varepsilon_0}, \qquad F = \frac{\sigma^2 A}{2\varepsilon_0}$$

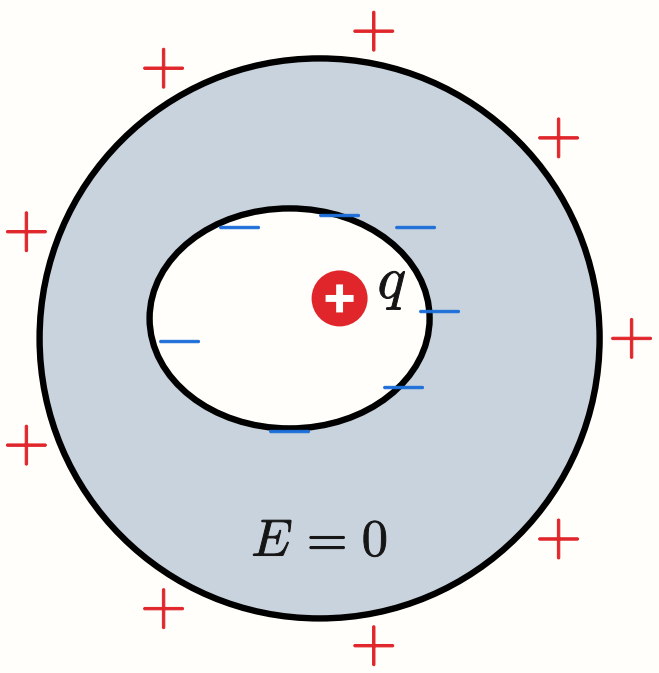
always OUTWARD, for either sign · the patch feels the average field $\sigma/2\varepsilon_0$

Example

$\sigma = 1\,\mu\mathrm{C\,m^{-2}}$: $E = 1.13 \times 10^5\,\mathrm{N\,C^{-1}}$, $P = 0.0565\,\mathrm{N\,m^{-2}}$.

Cavities and Shielding

1 Charge q inside a cavity of a conductor

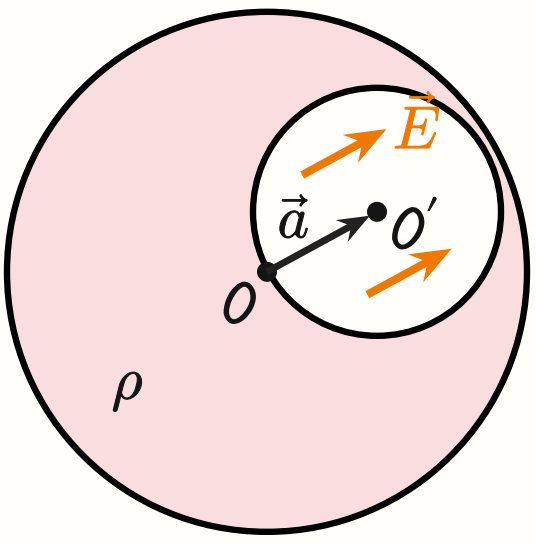


- Cavity wall: $-q$; outer surface: $+q$ (plus the conductor's own charge).
- Spherical outer surface: its charge is uniform wherever q sits.
- Uncharged conductor: outside $E = kq/r^2$ from the centre.

2 Electrostatic shielding

- EMPTY cavity: $E = 0$ inside, whatever charges or fields are outside.
- Uses: car or aircraft in lightning, metal box round circuits.
- It does NOT hide a charge inside from the outside, unless earthed.

3 Cavity in a uniformly charged sphere



$$\vec{E} = \frac{\rho\,\vec{a}}{3\varepsilon_0}$$

$\vec{a} = \overrightarrow{OO'}$ · UNIFORM in the whole cavity · full sphere + small sphere of $-\rho$

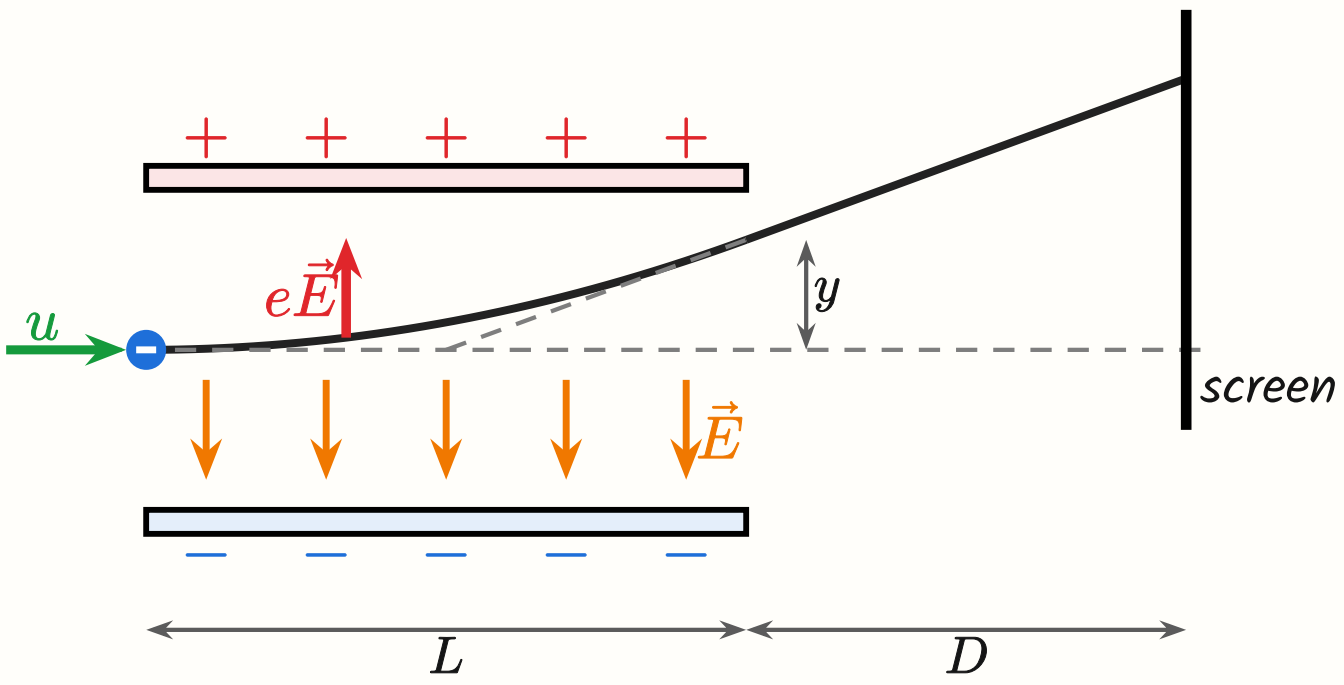
Charge in a Uniform Field

1 Released from rest

$$a = \frac{qE}{m}, \qquad K = qEd, \qquad v = \sqrt{\frac{2qEd}{m}}$$

uniform $\vec{E}$ · gravity neglected · d: distance moved along the force · $v \ll c$

2 Entering $\perp$ to $\vec{E}$ with speed u



$$y = \frac{qEx^2}{2mu^2}\ \text{(parabola)}, \qquad \tan\theta = \frac{qEL}{mu^2}$$

between the plates · on the screen: $Y = \tan\theta\,(L/2 + D)$

Example

Electron $(e/m = 1.76 \times 10^{11}\,\mathrm{C\,kg^{-1}})$, $u = 2 \times 10^{7}\,\mathrm{m\,s^{-1}}$, $E = 2 \times 10^{4}\,\mathrm{N\,C^{-1}}$, $L = 5\,\mathrm{cm}$: $y = 1.1\,\mathrm{cm}$, $\tan\theta = 0.44$.

Golden points

★ Electron bends to the + plate (force opposite to $\vec{E}$).

Common Mistakes

1 Exam traps

Avoid these

- ✗ Dipole $\vec{p}$ drawn from $+$ to $-$ (it is $-$ to $+$).
- ✗ $E_{ax} = 2kp/r^3$ used when r is not $\gg a$ (use the exact form).
- ✗ Equatorial $\vec{E}$ along $\vec{p}$ (it is opposite).
- ✗ Net force on a dipole in a NON-uniform field taken as zero.
- ✗ Gauss's law "fails" without symmetry (it holds; only finding E is hard).
- ✗ Outside charges ignored in $\vec{E}$ on a Gaussian surface (they add to E, not to the net flux).
- ✗ Sheet $\sigma/2\varepsilon_0$ used just outside a conductor (it is σ/ε_0).
- ✗ E inside a non-conducting solid sphere taken as zero (it is $\propto r$).
- ✗ θ in flux measured from the surface, not from the normal.
- ✗ Field line taken as the path of a charge.
- ✗ ε_r used when the medium does not fill all the space.

2 Recipe

Steps

- Point charges: superpose vectors, use symmetry.
- Continuous charge: dq, $d\vec{E}$, cancel $\perp$ parts, integrate.
- Symmetry (sphere / infinite line / infinite sheet): Gauss's law.

Summary

signs right? • conditions met? • vectors, not magnitudes